ROCK STRESS

In general, the distribution and magnitude of insitu stresses affect geometry, shape, dimensioning, excavation sequence and orientation of under ground excavations like caverns, tunnels etc.

The performance of hydraulic structures such as pressure tunnels is also strongly dependent on the magnitude and orientation of in-situ stresses since, in this case, hydraulic fracturing of the rock mass can be avoided.

ROCK STRESS

The orientation of long dimension of the cavern should be avoided perpendicular to the greater principal stress. The shape of the cavern should be selected to minimise the stress concentration especially in the region of high stresses. The layout of the complex should be planned so as to avoid crack propagation from one cavern to the other.

ROCK STRESS

Pressure tunnel, penstock and similar structures can be constructed and operated without lining if the in-situ stresses are greater than the internal water pressure.

In other situations like support design etc. the knowledge of the state of stresses can be integrated with design process along with the geological information.

ROCK STRESS

Stress fields alter the permeability of rock mass since compressive stresses tend to close natural fractures whereas tensile stresses tend to open them. On other hand, rock mass structures such as joints and foliation planes affect the distribution of in-situ stresses.

ROCK STRESS

The insitu stresses affect the rock properties in about twenty ways. Some of which are stated below:

- A stronger rock can sustain a higher in-situ stress.
- Stress concentrations decreases with displacements.
- In-situ stresses normal to discontinuities with large apertures will be low.
- Effective stress is reduced by increasing pore water pressure.
- Stress field alters permeability of rock mass

ROCK STRESS

- In-situ stress affects the stability of caverns.
- Hydrostatic in-situ stresses tend to close discontinuities.
- Stresses causes rock fracturing normal to minimum principal stress direction.
- High stress causes rock mass to fracture and its quality to deteriorate.
- Rock bursts in highly stressed rock masses affects excavation methods.
- Depth of tunnels/caverns is limited due to higher insitu stresses at depth.

ROCK STRESS

- Tunnels/caverns in highly stressed rock mass need more support.
- Ideal tunnel/cavern shape is controlled by in-situ stress field.
- Tectonic stresses, erosion, topography and other factors affect stress field.
- In-situ stress varies with depth.
- Discontinuities control magnitudes and directions of in-situ stress field.
- A highly variable in-situ stress field exists in a fractured rock mass.

ROCK STRESS

The rock stress field has both magnitude and orientation and can be considered to consist of two parts:

1. In situ stress field (Stress in field prior to excavation) ; and
2. Disturbance effects due to excavation.

Rock stress around an excavation = In situ stress field + Disturbance effects

ROCK STRESS

The In situ stress field primarily consists of two components:

1. Forces exerted by the weight of overlying rock mass; and
2. Large horizontal forces (tectonic forces) in the Earth's crust.

ROCK STRESS

The importance of rock stress and its influence on underground opening activity should be recognised and understood. The rock stress field around an excavation provides the driving forces that can cause rock instability of considerable violence. There are two types of stress measurements that can be undertaken:

1. **In situ rock stress measurements; and**
2. **Stress change measurements.**

ROCK STRESS

It is always advisable to measure it, in whatever best way possible.

There are several methods are can be used to estimate the magnitude and orientation of the rock stress field in terms of in situ stress levels or stress changes.

(A) IN SITU STRESS MEASUREMENT METHODS

- **CSIRO Hollow Inclusion cell (3D)**
- **Borehole slotter stressmeter (2D)**
- **USBM borehole deformation gauge (2D)**
- **Hydraulic fracturing method (2D)**
- **CSIR "doorstopper" (2D)**

ROCK STRESS

(B)STRESS CHANGE MONITORING METHODS

- **Vibrating wire stressmeter (1D)**
- **CSIRO Yoke gauge (2D)**
- **Seismic monitoring of a rock volume**
- **Flat or cylindrical pressure cell (1D)**

Methods for In-situ Stress Measurements

- Flat Jack Technique.
  - Surface
- Over Coring Method.
  - Shallow Depth
- Hydraulic Fracturing Technique.
  - Any Depth

EFFECTS OF IN-SITU STRESS ON ROCK PROPERTIES

- A highly variable in-situ stress field exists in a fractured rock mass.
- In-situ stresses tend to close discontinuities.
- In-situ stresses normal to discontinuities with large apertures will be low.
- A stronger rock can sustain a higher in-situ stress.
- High stress causes rock mass to fracture and its quality to deteriorate.

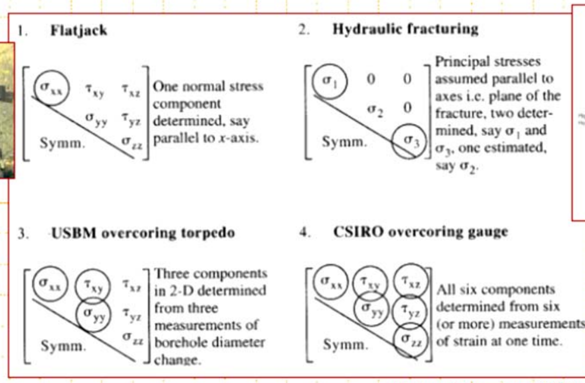
Effects of In-situ Stress on Rock Properties

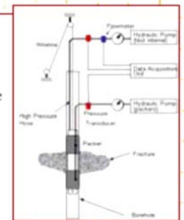
- Effective stress is reduced by increasing pore water pressure.
- In-situ stress varies with depth.
- Stress field alters permeability of rock mass.
- Stresses cause rock fracturing normal to minimum principal stress direction.
- Discontinuities control magnitudes and directions of in-situ stress field.
- Tectonic stresses, erosion, topography and other factors affect stress field.

In-situ Stress Measurement Techniques

- Most widely used techniques are:
  - Flat jack method
  - Overcoring method using USBM Gauge & CSIRO HI Cell
  - Borehole slotting technique
  - Hydraulic Fracturing Method
- However hydraulic fracturing method is most suitable for shallow and deep depths.

Methods of Stress Determination



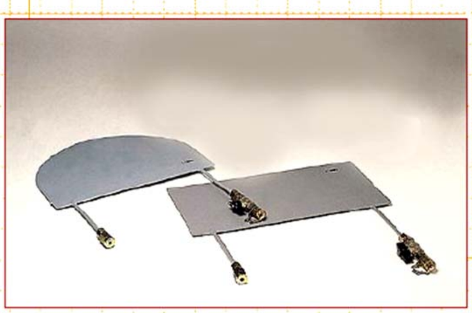


Hudson & Harrison (1997)

… the four ISRM suggested methods for rock stress determination and their ability to determine the 6 independent components of the stress tensor over one test/application of the particular method.

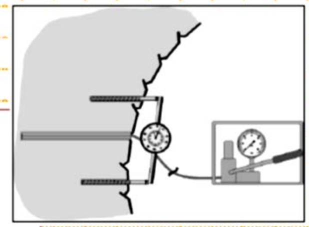
Flatjack Method

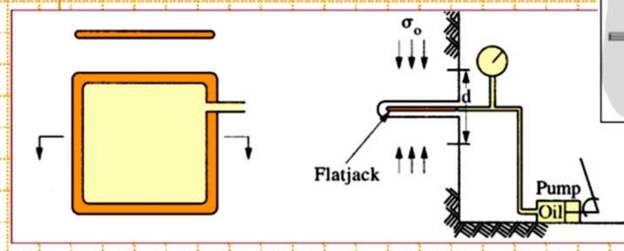
A flatjack is comprised of two metal sheets placed together and welded around their periphery. A feeder tube inserted in the middle allows the flatjack to be pressurized with oil or water.



Flatjack Method

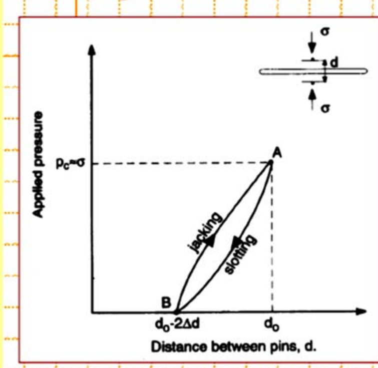
The **flatjack** method involves the placement of **two pins fixed** into the wall of an excavation. The distance, *d*, is then measured accurately. A **slot is cut** into the rock between the pins. If the normal stress is compressive, the pins will move together **as the slot is cut**. The flatjack is then placed and grouted into the slot.

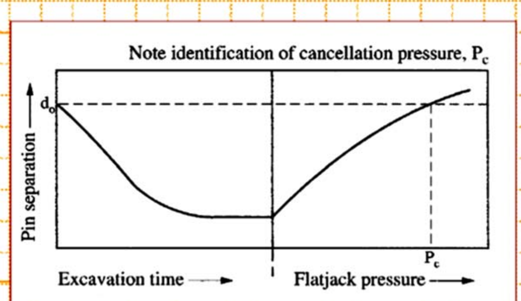




Flatjack Method

On pressurizing the flatjack, the pins will move ***apart****. It is assumed that, when the pin separation distance reaches the value it had* ***before*** *the slot was cut, the force exerted by the flatjack on the walls of the slot is the same as that exerted by the pre-existing normal stress.*

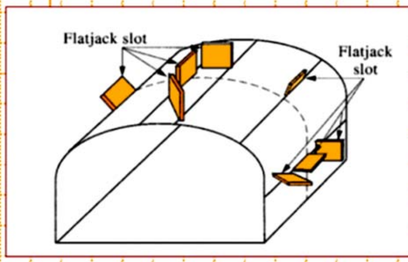




Hudson & Harrison (1997)

Flatjack Method

The major disadvantage with the system is that the necessary minimum number of 6 tests, at different orientations, have to be conducted at 6 different locations and it is therefore necessary to distribute these around the boundary walls of an excavation.



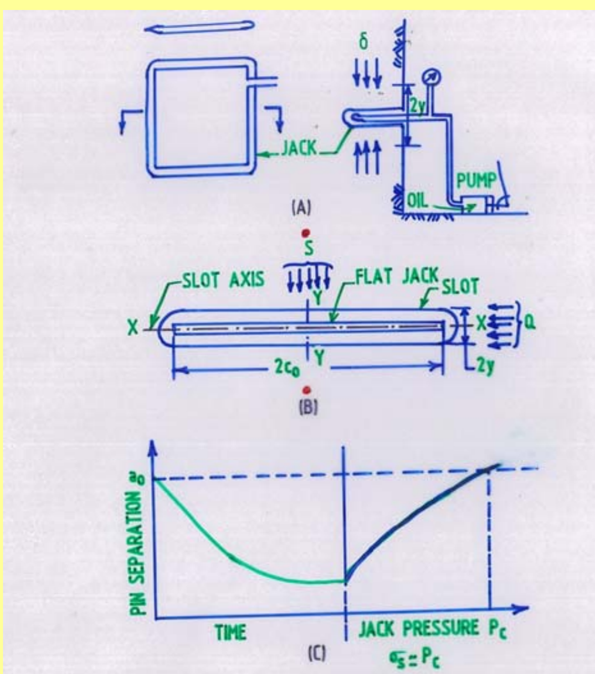
1. **Flatjack**

$$\begin{bmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ & \sigma_{yy} & \tau_{yz} \\ \text{Symm.} & & \sigma_{zz} \end{bmatrix}$$

One normal stress component determined, say parallel to x-axis.

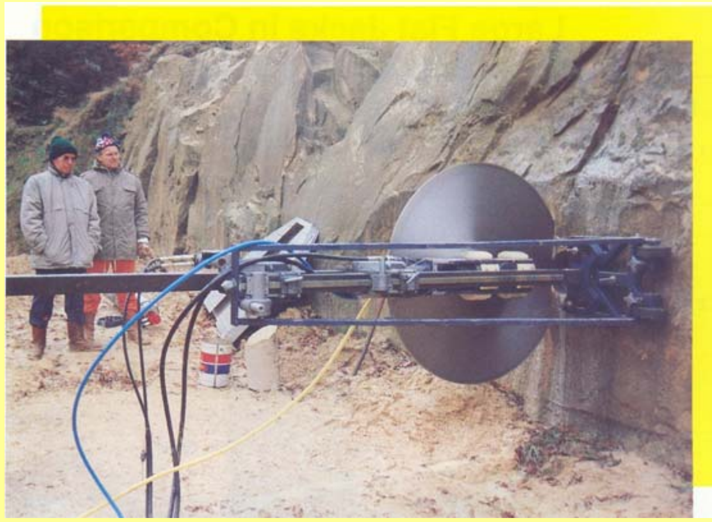
Hudson & Harrison (1997)

It is also important to note that the excavation from which the tests are made will disturb the pre-existing stress state, and so the new redistribution of stresses should be accounted for.



Flat Jack Test Set-up

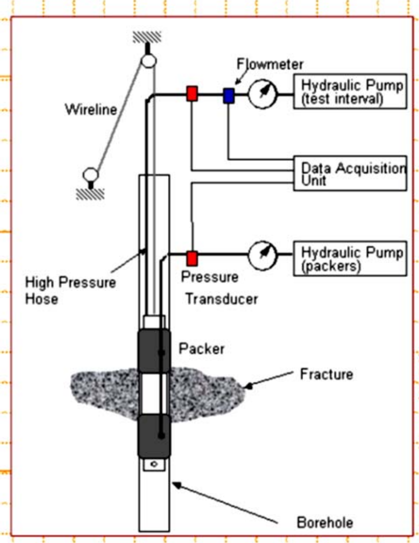
Extra Large Flat Jack Test



For Dam and Underground Construction in Rock

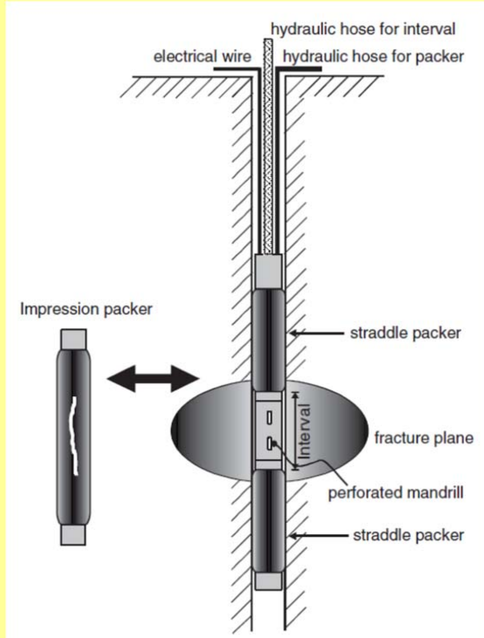
Hydraulic Fracturing Method

The **hydraulic fracturing method** involves the pressuring of a borehole interval, typically 1 m long, isolated using a straddle packer system. The isolated zone is **pressurized** by the hydrofrac fluid until a fracture occurs in the rock.



Haimson & Lee (1984)

Schematic diagram of hydrofracturing test.



Equipment and procedure
In the hydrofracturing test, a certain interval in borehole is completely sealed off by straddle packer (Figure), and then this interval is pressurized until tensile crack starts to generate.

When the rock mass is isotropic, the hydrofracturing tensile cracks will be generated to the plane perpendicular to the horizontal minimum principal stress in which resistance to the tensile failure is lowest.

When the pumping into the interval continues after hydrofractures are generated, the pressure of the interval does not increase and reaches to a breakdown pressure.

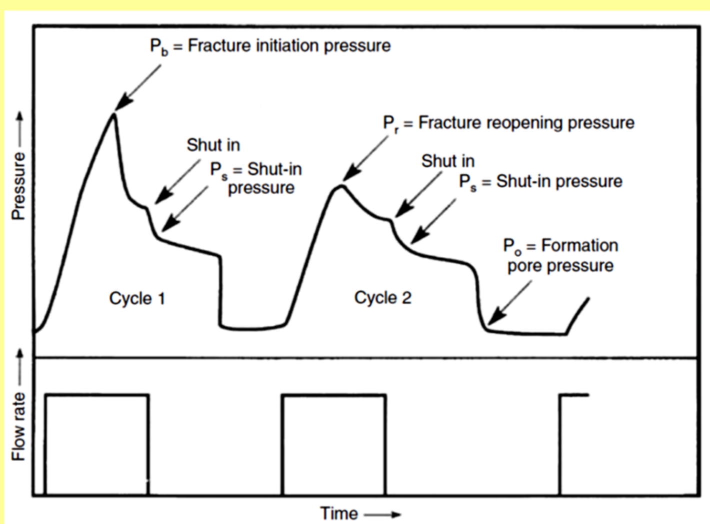
An idealized hydrofracturing pressure history is shown in Figure, and the horizontal maximum and minimum principal stresses are obtained from the equations for the state of stress around a circular opening.

From the hydrofracturing pressure history curves, it is possible to determine the fracture initiation (breakdown) pressure PC, the shut-in pressure PS and the fracture reopening pressure Pr.

The shut-in pressure is the pressure at which a hydrofracture stops propagating and closes following pump shut-off, so it is the pressure for maintaining the balance with the minimum principal stress acting perpendicularly on the fracture plane.

That is, the shut-in pressure equals the horizontal minimum principal stress in the test site.

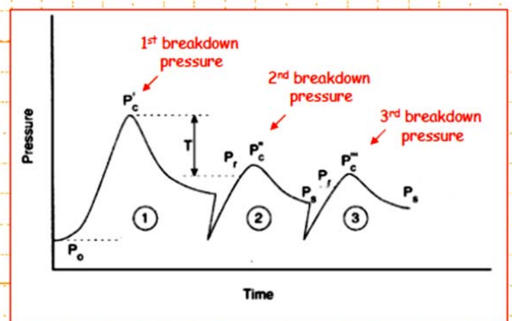
$$\sigma_h = P_S$$



Idealized hydrofracturing pressure history curves (Amadei & Stephansson, 1997).

Hydraulic Fracturing Method

The two measurements taken are the water pressure when the fracture occurs and the subsequent pressure required to hold the fracture open. These are referred to as the **breakdown pressure** (P_c' or P_B) and the **shut-in pressure** (P_s).

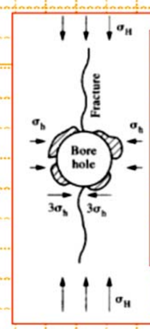


Amadei & Stephansson (1997)

Hydraulic Fracturing Method

In calculating the *in* situ stresses, the **shut-in pressure** (P_s) is assumed to be equal to the **minor horizontal stress**, σ_h.

The **major horizontal stress**, σ_H, is then found from the **breakdown pressure** (P_c' or P_B). In this calculation, the breakdown pressure has to overcome the minor horizontal principal stress (concentrated three times by the presence of the borehole) and overcome the *in situ* tensile strength of the rock; it is assisted by the tensile component of the major horizontal principal stress.



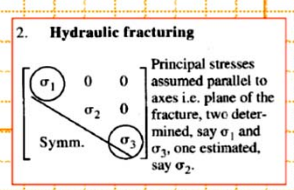
Relationships

$\sigma_h = P_s$

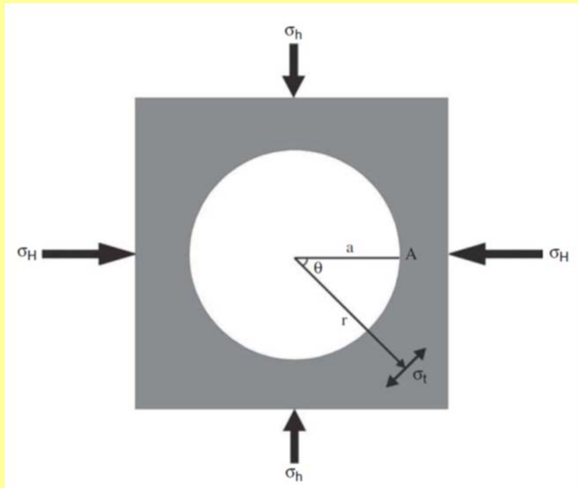
$\sigma_H = 3\sigma_h - P_c' - P_o + \sigma_t$

$\sigma_t = P_c' - P_r$

$\sigma_H = 3\sigma_h - P_r - P_o$



Hudson & Harrison (1997)



State of stresses around a circular opening (Jaeger et al., 2007)

The tangential stress σt at a point in a distance of r from the origin of a circular opening of radius a is given by Kirsch solution (Jaeger et al., 2007) and described below in the following equation.

$$\sigma_t = \frac{1}{2}(\sigma_H + \sigma_h)\left(1 + \frac{a^2}{r^2}\right) - \frac{1}{2}(\sigma_H - \sigma_h)\left(1 + 3\frac{a^4}{r^4}\right)\cos(2\theta)$$

where

σ_H; maximum horizontal principal stress,
σ_h; minimum horizontal principal stress,
σ_t; tangential compressive stress at a point in a distance of r from borehole axis,
a; radius of borehole.

Increasing the pressure of the interval, a crack starts to propagate at point A and the tangential compressive stress at this point becomes to be a minimum. At point A, r = a and θ = 0, so the above equation can be expressed as.

$$\sigma_t = 3 \cdot \sigma_h - \sigma_H$$

The conditions for a new, vertical tensile crack are that the tensile stress at point A should become equal to the tensile strength –T0. Applying this to the hydrofracturing experiment yields as a condition for creation of hydrofractures

$$3 \cdot \sigma_h - \sigma_H - P_C = -T_0$$
$$T_0 = P_C - P_r$$

The above equations allow the maximum horizontal principal stress to be determined as

$$\sigma_H = 3 \cdot \sigma_h - P_r$$

Also the pore pressure into the rock mass around the borehole should be considered. Therefore, the above equation can be expressed as

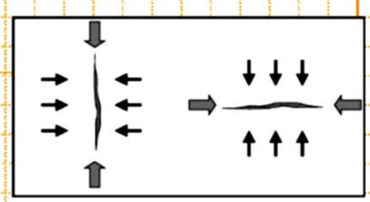
$$3 \cdot \sigma_h - \sigma_H - P_C = -T_0 + P_0$$
$$\sigma_H = 3 \cdot \sigma_h - P_r - P_0$$

Hydraulic Fracturing Method

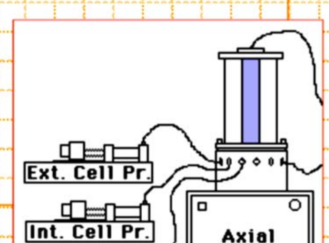
The analysis assumes that the induced fracture has propagated in a direction perpendicular to the minor principal stress.

Other assumptions include that the rock behaves elasticity (from which the borehole stress concentration factor of three is derived), and is impermeable (i.e., the pumped water has not significantly penetrated the rock and affected the stress distribution).

The tensile strength of the rock can be obtained from test performed by pressurizing hollow rock cylinders.



(a) $K_0 < 1$ Vertical fracture   (b) $K_0 > 1$ Horizontal fracture

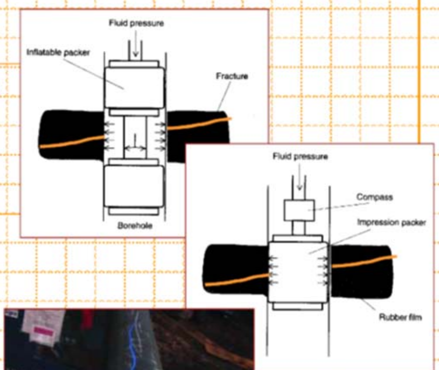


Hydraulic Fracturing Method

There are also several problems inherent in the use of this method. For example, it can often be difficult, if not impossible, to identify a 1 m length of borehole which is fracture free.

Furthermore, there can be difficulties measuring water pressures accurately, and in correctly identifying the breakdown and shut-in pressures.

Lastly, it is often an oversimplifying assumption that the borehole is parallel to a principal stress.

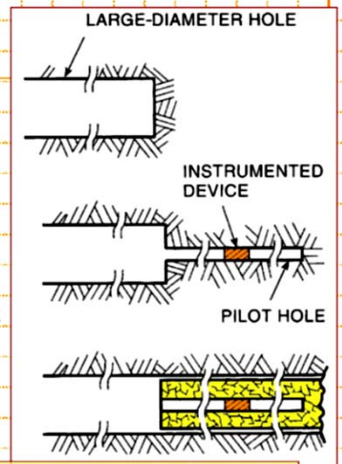


Overcoring Method

First, a large diameter borehole is drilled (between 60 and 220 mm) to a sufficiently large distance so that stress effects due to any excavations can be neglected.

Second, a small pilot hole (e.g. 38 mm) is drilled. The measuring device is then inserted and fastened in this hole.

Thirdly, the large diameter hole is resumed, relieving stresses and strains in the hollow rock cylinder that is formed. Changes in strain are then recorded with the instrumented device as the overcoring proceeds past the plane of measurement.



Following overcoring, the recovered overcore (containing the instrumented device) is then tested in a biaxial chamber to determine the elastic properties of the rock.

Stress Determination Methods - Summary

Eberhardt & Stead (2011)

Method	Advantages	Limitations	Suitability
Overcoring	Most developed technique in both theory and practice; 3-D.	Scatter in data due to small rock volume; requires drill rig.	Measurement depths down to 1000m.
Doorstopper	Works in jointed and high stressed rocks.	Only 2-D; requires drill rig.	For weak or high stressed rocks.
Undercoring	Simple measurements; low cost; can utilize existing underground excavation.	Measures local stresses (must be related to far-field stresses); rock may be disturbed.	During excavation.
Hydraulic fracturing	Can utilize existing boreholes; tests large rock volume; low scatter in results; quick.	Only 2-D; theoretical limitations in the evaluation of s_H.	Shallow to deep measurements.
HTPF	Can utilize existing boreholes; 3-D; can be applied when high stresses exist and overcoring and hydraulic fracturing fail.	Time-consuming; requires existing fractures in the hole with varying strikes and dips.	Where both overcoring and hydraulic fracturing fail.

Stress Determination Methods – Summary (cont.)

Eberhardt & Stead (2011)

Method	Advantages	Limitations	Suitability
ASR/DSCA/ RACOS	Usable for great depths.	Complicated measurements on the micro-scale; sensitive to several factors	Estimation of stress state at great depth.
Acoustic emissions (Kaiser effect)	Simple measurements.	Relatively low reliability; requires further research.	Rough estimations.
Focal mechanisms	For great depths; existing information from earthquake occurrence.	Information only from great depths.	Seismically active areas.
Core discing	Information, obtained from borehole drilling.	Only qualitative estimation.	Stress estimation at early stage.
Borehole breakouts	Existing information obtained at an early stage; relatively quick.	Orientation information only; theory needs to be further developed to infer stress magnitudes.	Deep boreholes or around deep excavations.
Back analysis	High certainty due to large rock volume.	Theoretically, not unique solution.	During excavation.
Geological indicators	Low cost; 2-D/3-D.	Very rough estimation; low reliability.	At early stage of project.

ENGINEERING PROPERTIES OF ROCK MASS

The extent to which ground can be expected to move is primarily dependent on the mechanical properties / engineering characteristics of the rock / ground mass. Consequently, the excavation operator of a site will need to determine all the mechanical properties of the rock mass that are relevant to specific mechanisms of ground movement or failure expected at that site.